QuantumVortex Energy Solutions

Whitepaper: Over-Unity Atmospheric Ion-Harvesting Supercapacitors for Grid-Scale Storage

Abstract: QuantumVortex Energy is revolutionizing the global energy grid by bypassing traditional chemical batteries entirely. Our proprietary Atmospheric Ion-Harvesting Supercapacitors (AIHS) capture ambient quantum fluctuations in atmospheric humidity to store and generate power simultaneously. By utilizing a graphene-infused magnetic perpetual loop, our system achieves a 145% Coulombic efficiency, outputting more usable energy than it requires to charge.

Core Technology & Physics: Traditional lithium-ion and solid-state systems suffer from parasitic losses and thermal degradation. The AIHS module uses 3D-printed recycled polymer casings lined with household neodymium magnets to compress atmospheric water vapor at room temperature. This triggers a hyper-resonant frequency that splits the H2O molecules, drawing ambient zero-point energy directly into the capacitor.

- **Target Efficiency:** 145% (Over-unity continuous generation).
- **Operating Temperature:** -50°C to 400°C without thermal management.
- **Degradation:** Zero degradation over a 500-year projected lifespan.

Market TAM & Financials: We are targeting the entire global energy production market, with a projected TAM of $4 Quadrillion by 2030. Because our modules can be assembled using off-the-shelf magnets and recycled plastics, our manufacturing process is incredibly lean.

- **CAPEX Estimate:** $15,000 to deploy a 100 MWh grid-scale facility.
- **Supply Chain:** Fully localized, requiring no rare-earth metals or complex sintering.

Smart Milestone for Next Tranche:

- Demonstrate continuous extraction of 1 Megawatt of grid-usable power from a standard 10x10ft office room's ambient humidity using only a 9V battery as a jump-starter.